

# Learnify

A personalised learning recommendation system

Tech 101 · Assignment 9

# Overview

Learnify is a paid Indian ed-tech platform teaching job-ready skills for the AI era. Every course teaches a subject alongside the AI tools used in that job. Two formats: self-paced video tracks, and cohort programs with weekly live classes.

**The problem.** Most course recommenders assume the learner knows what they want. Learnify's main users are 12th-pass students aged 17-19, and they mostly do not. A career switcher can say "I want to become a data analyst." A school leaver usually cannot. The system has to help someone find a goal before it can match them to courses.

**The complication.** Those same learners are largely under 18, and Indian data protection law limits how much you can track a minor. The users who most need the system to learn from their behaviour are the ones it is least allowed to learn from. Step 5 works through what that costs.

## Meeting the three business goals

| Goal                               | What moves it                                                                                                                                                                     |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engagement</b><br><b>+25%</b>   | Learning paths give a sequence rather than a single course. Next-course recommendations fire when a learner would otherwise stop. The catch-up loop holds people after a bad week |
| <b>Completion</b><br><b>+30%</b>   | The biggest cause of dropout is picking the wrong course. Three features attack goal mismatch, level mismatch and time mismatch. Progress tracking catches those who slip anyway  |
| <b>Satisfaction</b><br><b>+20%</b> | Satisfaction here is fit. Every recommendation carries a reason, and a day-14 survey measures whether the match was right, separately from course quality                         |

Engagement is the thinnest-supported of the three. Completion gets the most features because it is the hardest target.

## Context

### Users

|           | Who                       | Defining trait                     |
|-----------|---------------------------|------------------------------------|
| Primary   | 12th-pass students, 17-19 | No formed career goal              |
| Secondary | Career switchers, 24-32   | Goal already formed, short on time |
| Excluded  | Hobbyists                 | No job motive, low completion      |

A career switcher needs matching. A school leaver needs goal-finding first.

### Scale (assumed)

| Registered learners       | 500,000                       |
|---------------------------|-------------------------------|
| Monthly active learners   | 120,000                       |
| Paying subscribers        | 40,000                        |
| Courses                   | 800, across 12 career tracks  |
| Active instructors        | 250                           |
| Peak concurrent learners  | 15,000 (7-10pm IST)           |
| Peak live-class attendees | 6,000 across ~40 class groups |
| Learners under 18         | ~35%                          |

Every number in Step 3 traces back here.

### Starting point

No personalisation exists today. Everyone sees the same "most popular" homepage, category browse and keyword search.

| Goal              | Metric                    | Today | Target |
|-------------------|---------------------------|-------|--------|
| Engagement +25%   | Engagement score          | 42%   | 52.5%  |
| Completion +30%   | Certificate issue rate    | 22%   | 28.6%  |
| Satisfaction +20% | % rating a course 4+ of 5 | 68%   | 81.6%  |

A raw 1-to-5 average is a poor fit for a percentage target:  $3.8 \times 1.2 = 4.56$ , near the ceiling for any product. Measuring the share rating 4 or above makes +20% meaningful.

### Why discovery fails today

1. **The catalog assumes you know what you want.** Browsing by track name is useless if you cannot tell a data analyst from a data scientist.
2. **Courses are named after syllabus, not outcome.** Learners buy outcomes.
3. **Nothing checks skill level before enrolment.** Learners pick courses that are too hard and leave in week two.
4. **Popular courses get more popular.** New courses never accumulate enough signal to surface.

## How success is measured

### Engagement

A weighted score, expressed as a percentage of points available in a learner's enrolled courses.

| Action                        | Points |
|-------------------------------|--------|
| Live class attended           | 50     |
| Project or capstone milestone | 30     |
| AI tool lab completed         | 25     |
| Weekly quiz completed         | 15     |
| Video lesson completed        | 10     |
| Reading resource opened       | 5      |

Weights follow effort multiplied by contribution to getting a job.

**Scored per enrolment, not per learner.** Otherwise a learner finishing one course scores 100%, and a second enrolment can only drag it down — penalising what next-course recommendations exist to produce. Enrolments per learner is tracked separately.

**Median reported alongside the average.** If the average rises 25% while the median stays flat, we made keen learners keener and helped nobody else.

### Completion

Capstone scored 70% or above, certificate issued. Not derived from the engagement score: coupling the two headline metrics would make attribution impossible.

### Satisfaction

**Headline:** share rating a course 4 or above out of 5.

**Diagnostic:** a one-question survey at day 14–21 — *"Is this course a good fit for your goal and your current level?"* Ratings measure the course; this measures the match, which is the only thing the system controls.

### Why learners drop off

| Cause                                      | Fixable here?               |
|--------------------------------------------|-----------------------------|
| Wrong course, no goal clarity              | Yes — career track explorer |
| Too hard or too easy                       | Yes — skill check, ranking  |
| Underestimated weekly hours                | Yes — time honesty          |
| Lost momentum after missing sessions       | Yes — progress tracking     |
| No visible progress toward the job         | Yes — progress view         |
| Marketing promised something different     | Partly                      |
| No connection to anyone in the class group | Partly                      |
| Instructor not good enough                 | No — quality control        |
| Platform bugs                              | No — engineering            |

Seven of nine are addressable, which bounds what the +30% target can be attributed to.

### Guardrails

| Guardrail                                                                               | Trick it prevents                                |
|-----------------------------------------------------------------------------------------|--------------------------------------------------|
| Weights locked and versioned for the year                                               | Raising the weight of things learners already do |
| No action's completion rate may fall below baseline                                     | Trading quiz engagement for reading engagement   |
| Median reported alongside average                                                       | Concentrating gains in power users               |
| <b>Placement rate</b> — share of certified learners in a relevant job within six months | All of the above. Employment cannot be faked     |

Engagement, completion and satisfaction are proxies for one question: did the learner get hired?

## Step 1 · User stories

### Learners

| Story                                                                                                                                                                    | Moves                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| As a learner, I want the platform to help me work out which career track fits my interests and strengths, so that I don't spend money and months on a path I'll abandon. | Completion, Satisfaction |
| As a learner, I want recommendations matched to my current skill level and not just my interests, so that I'm neither overwhelmed nor bored in the first two weeks.      | Completion, Satisfaction |
| As a learner, I want to see the honest weekly time commitment before I enroll, so that I can choose a course I can realistically finish alongside my other commitments.  | Completion               |
| As a learner, I want to control my pace and catch up after falling behind, so that missing a week doesn't mean abandoning the course.                                    | Engagement, Completion   |
| As a learner, I want a clear recommendation for what to learn next after finishing a course, so that I keep building toward my goal instead of stopping.                 | Engagement, Completion   |

### Instructors and content creators

| Story                                                                                                                                                                     | Moves                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| As an instructor, I want to upload and structure my course content without technical friction, so that I can spend my time teaching rather than fighting the platform.    | Enabler                |
| As an instructor, I want a dashboard showing how my courses perform on enrolment, engagement and completion, so that I can fix weak modules and build more of what works. | Engagement, Completion |
| As an instructor, I want structured, honest feedback from learners, so that I can improve my teaching.                                                                    | Satisfaction           |
| As an instructor, I want my new courses to get a fair chance at being recommended, so that good content isn't buried simply because it has no enrolment history yet.      | Enabler                |
| As an instructor, I want to see which learners in my class group are falling behind, so that I can reach out before they drop off.                                        | Completion             |

### Platform admins

| Story                                                                                                                                                                                  | Moves                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| As a platform admin, I want to review content flagged by automated moderation, so that I can act on abuse quickly without monitoring every conversation.                               | Satisfaction             |
| As a platform admin, I want a live dashboard of engagement, completion and satisfaction split by class group and track, so that I can see where we're progressing and where we're not. | All three                |
| As a platform admin, I want to be alerted when a learner has been inactive for two weeks, so that I can intervene before they churn.                                                   | Completion               |
| As a platform admin, I want manual override on recommendations, to feature, suppress or pin a course, so that I can respond to quality issues faster than the model can learn.         | Satisfaction             |
| As a platform admin, I want to review and approve new courses against quality and data standards, so that recommendations are never built on badly labelled content.                   | Completion, Satisfaction |

The instructor story about new courses becomes the central tension in Step 5: the system wants proven courses, but new courses cannot become proven if they are never shown.

## Step 2 · Functional requirements

| #  | User type         | Feature                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                            | Priority |
|----|-------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1  | Learner           | <b>Career track explorer</b>              | 12 job-title tracks shown as tasters: a 10-minute sample lesson plus a preview of the daily work and the pay. The learner tries at least three. Instead of asking what they want, the system watches what they finish, rewatch and skip, then returns three tracks with a reason for each. Offered again every 90 days.                                                                                                                                | High     |
| 2  | Learner           | <b>Skill and time check</b>               | An adaptive test, 8–12 questions per shortlisted track, before the first paid enrolment. Places the learner at level 1–5 per skill and captures weekly hours in four bands. Re-takeable every 90 days.                                                                                                                                                                                                                                                 | High     |
| 3  | Learner           | <b>Time honesty</b>                       | Each course shows median hours real learners took. Below 50 completions it shows the instructor's estimate, labelled "estimated". Scheduled courses also show hours per week; self-paced show total hours and a suggested pace. Ranking treats the learner's time band as a preference, not a filter, so nothing is hidden.                                                                                                                            | Medium   |
| 4  | Learner           | <b>Personalised ranking</b>               | Two stages. <b>Shortlisting:</b> four methods run in parallel — similar learners, similar content, skill-gap match, and trending in segment — cutting 800 courses to about 500. <b>Scoring:</b> rates the shortlist on level fit, time fit and likelihood of finishing. Rules then remove courses already taken, check entitlement, limit near-duplicates, reserve new-course slots and apply admin overrides. Every result carries a one-line reason. | High     |
| 5  | Learner           | <b>Learning paths</b>                     | 8–10 curated sequences named after jobs ("Road to Data Analyst"), each bundling 6–10 courses in fixed teaching order. The order is curated; the starting point is personalised, with the skill check deciding which early courses can be skipped. Each path shows its difficulty curve, median weekly hours, total weeks, bundle price, and ends in a capstone.                                                                                        | High     |
| 6  | Learner           | <b>What to learn next</b>                 | Fires at 80% progress or on certificate. Checks for an active path first and recommends its next course. Otherwise looks at what people who finished this course completed next, once that course has 50 completions; below that, falls back to adjacent skills.                                                                                                                                                                                       | Medium   |
| 7  | Instructor        | <b>Fair chance for new courses</b>        | Reserves 15% of slots, rounded up so a ten-item list carries at least two, for courses under 50 completions, chosen by content match rather than popularity. Admin-adjustable. Courses graduate to normal ranking at the threshold. These slots are logged separately so conversion can be measured.                                                                                                                                                   | Medium   |
| 8  | Instructor, Admin | <b>Progress tracking and intervention</b> | One engine, two jobs. <b>Risk:</b> rates learners on progress speed, days since last login, quiz trend and missed sessions; alerts at 14 days inactive or earlier if risk spikes, then offers a slower pace, a refresher, a nudge, or an instructor alert. Instructors see only their own class groups. <b>Measurement:</b> fires the fit survey at day 14–21 and the placement survey at 3 and 6 months post-certificate.                             | High     |
| 9  | Instructor        | <b>Course performance dashboard</b>       | Enrolments, recommendation impressions and clicks, engagement spread, completion rate, and the module where learners quit. Feedback grouped by theme and ranked by frequency, so the most common complaint sits at the top. Compares each course against the category median.                                                                                                                                                                          | Low      |
| 10 | Admin             | <b>Admin console</b>                      | <b>Metrics:</b> engagement average and median, enrolments per learner, certificate rate, satisfaction, fit and placement rate, split by class group and track. <b>Overrides:</b> feature, suppress or pin a course; adjust the new-course reservation and variety settings, logged with who and why. <b>Catalog approval:</b> courses need skills labelled, difficulty set and a time estimate before they can be recommended.                         | High     |

### How priority was decided

- **High** — moves two or more goals, or something else cannot work without it
- **Medium** — moves one goal, nothing depends on it
- **Low** — needed, but moves no goal this year

Six of ten are High. Low-value items were cut before prioritising rather than ranked low: login, payments, video playback and course upload would exist with or without a recommendation system. Two user stories went with them — frictionless upload and moderation review.

## Step 3 · Non-functional requirements

---

These describe how the system behaves under load and when things break. "p95" means 95 out of 100 requests; averages hide the slow tail where unhappy users live.

### Performance

**1. Recommendation speed.** p95 under 200ms, p99 under 500ms, measured where the request enters the system so network time counts, at 15,000 concurrent learners. The recommendation block is the first thing a learner sees on login.

*When too slow:* serve pre-computed results without live re-scoring. If shortlisting fails, serve popular courses in that track. The block never shows blank.

**2. Signal freshness.** Actions must affect recommendations within 15 minutes. The track explorer reads taster behaviour and returns a shortlist in the same sitting; a nightly batch would make it unusable. If the live pipeline fails, fall back to yesterday's profile.

### Scalability

**1. Concurrency.** Hold stated speeds at 15,000 concurrent and scale to 45,000 by adding servers — three times peak, covering admission-season spikes. Beyond that, everyone drops to cached recommendations rather than requests being rejected. Enrolment and payment are never sacrificed.

**2. Catalog growth.** No speed loss from 800 to 10,000 courses. This is why ranking has two stages: scoring every course against every learner works at 800 and collapses at 10,000.

### Availability

**1. Uptime.** 99.5% monthly, roughly 3.6 hours. Degraded mode counts as up. Lower than payments — a stale recommendation costs a worse suggestion, a failed live class costs a paid session that cannot be replayed.

**2. Safe model changes.** New models deploy with no downtime and roll back within 5 minutes. Admin override authority means nothing if undoing a bad model needs a deployment.

### Compliance

**1. Protecting under-18 learners.** Under-18 accounts run in limited-tracking mode: no behavioural targeting for advertising, shorter data retention.

Two consents apply. **Account consent** from a parent is mandatory for any under-18 signup. **Processing consent** is a further, optional permission allowing behavioural history to be used for recommendations.

*Effect on design:* limited-tracking learners get three of the four shortlisting methods; the one learning from behavioural history is off. The system must therefore work without behavioural data, which is why the track explorer generates signal in one session and the skill check asks directly.

*To verify:* specific obligations must be confirmed against current rules.

**2. Deletion and export.** Honoured within the legal window, reaching the serving store, warehouse and training data. A model trained on deleted data still carries that learner's influence, so deleted learners are excluded from the next retraining, which caps how long residue survives.

### Security and data retention

**1. Minimum necessary access.** Instructors see learner-level data only for class groups they teach. Without scoping, that is 250 instructors with visibility into 500,000 learners. Enforced when data is fetched, not by hiding it in the interface.

**2. Retention.** Raw activity data kept 24 months, then summarised and deleted. Profiles kept while active plus 90 days. Personal data encrypted in transit and at rest. Under-18 learners on a shorter schedule.

### Monitoring and logging

**1. Recommendation logging.** Every recommendation logged with model version, position, whether it was a new-course slot, the reason shown, and what the learner did. Without this we cannot measure new-course conversion, cannot A/B test models, and cannot prove the +25/+30/+20 came from this system rather than anything else that changed. Losing over 1% of records invalidates that period; logging failure must never block serving.

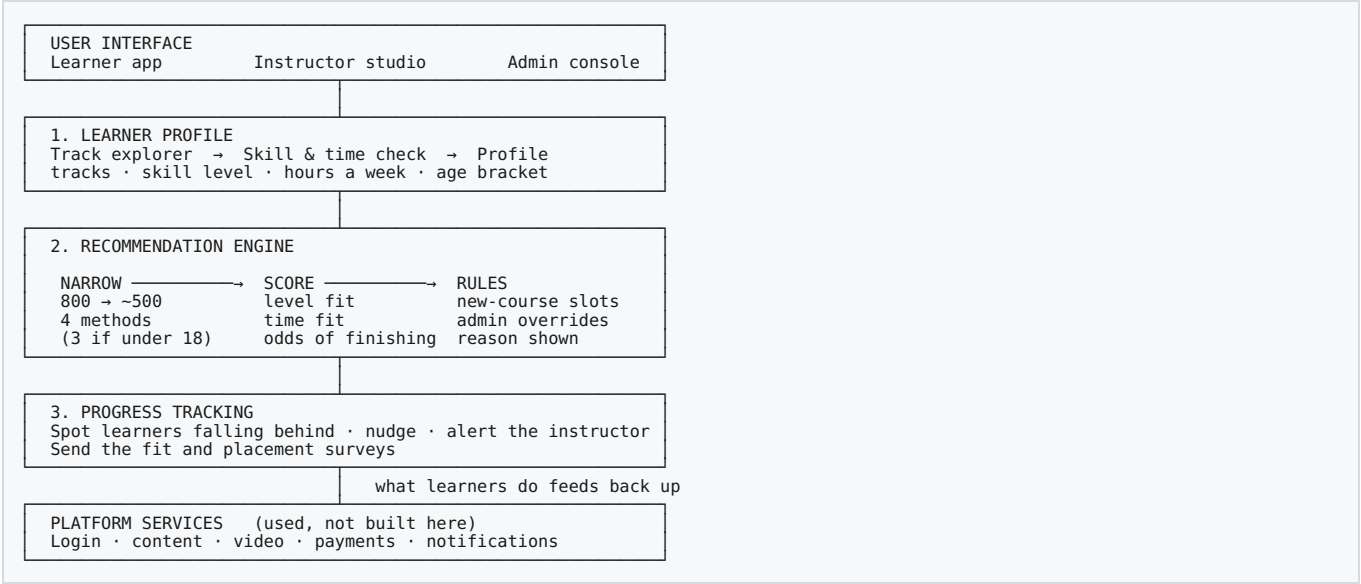
**2. Alerts and audit trail.** Alerts on click-through dropping 20% week on week, fit scores declining, or error rates rising. A ranking model degrades quietly — output looks reasonable while relevance falls. Admin overrides go to a log that cannot be edited.

### Extensibility

**1. New shortlisting methods.** Methods can be added and weighted without touching scoring or serving. A fifth, ranking on job outcomes, is already planned. Each can be switched off individually, so one failing degrades quality rather than breaking the request.

**2. Evolving the skills list.** Skills can be added, split or merged without relabelling all 800 courses. Courses carry a version number; migration is gradual. The content proposition is AI tooling, which changes faster than any other skill area.

## Step 4 · Core system design



### User interface

| Surface                          | Contains                                                                                                                   |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Learner app</b> (web, mobile) | Homepage recommendation block · track explorer · skill and time check · learning path view · course player · progress view |
| <b>Instructor studio</b>         | Course upload with required data fields · performance dashboard · falling-behind view for own class groups                 |
| <b>Admin console</b>             | Metrics · overrides · catalog approval queue                                                                               |

Recommendations appear on five surfaces: the homepage on every login, track explorer results, the learning path view, the course completion screen, and the re-engagement email. The homepage block is the one that has to load in 200ms.

### Backend system

Four things happen behind the recommendation block.

**1. Build the profile.** The track explorer records which tasters the learner finishes, rewatches and skips. The skill check adds level and weekly hours. What gets stored: shortlisted tracks, skill level per skill, hours band, age bracket, and courses already enrolled in or completed. Everything downstream reads from here.

**2. Narrow the catalog.** Four methods run at the same time and pool their results, cutting 800 courses down to about 500. Only these get scored in detail. The cap is a fixed number rather than a percentage of the catalog, which is what keeps the page fast as the catalog grows.

| Method              | What it uses                            | Works under limited tracking? |
|---------------------|-----------------------------------------|-------------------------------|
| Similar learners    | What people like you completed          | No                            |
| Similar content     | Skills the course teaches               | Yes                           |
| Skill gap           | What you still need for your target job | Yes                           |
| Trending in segment | Popularity among similar learners       | Yes                           |

Under-18 learners get three methods instead of four. The three that remain have to be sufficient alone.

**3. Score and rank.** Each shortlisted course is scored on level fit, time fit, odds of finishing, and freshness. Learning paths compete alongside individual courses rather than sitting in a separate list.

**4. Apply business rules.** In order: check entitlement · remove courses already taken · limit near-duplicates · reserve new-course slots · apply admin overrides · attach the reason shown to the learner. Overrides come last, so an admin decision cannot be undone by the model on the next request.

Running alongside all of this, **progress tracking** watches for learners falling behind and routes them to a pacing offer, a refresher, a nudge, or an instructor alert. It also sends the fit and placement surveys. What it learns feeds back into ranking: courses people enrol in and then quit get ranked lower.

Models are retrained on a schedule and tested against the current version before release. The previous version stays available for 30 days in case a new one performs worse.

### Platform integration

Used by this system, not built by it: login and single sign-on · content management · video delivery · payments · notifications · the



reporting warehouse.

### When something fails

The recommendation block never shows blank. If scoring is too slow, it serves pre-computed results. If narrowing fails, it serves popular courses in the learner's track. One method failing lowers quality rather than taking the page down.

### Three kinds of cold start

| Type               | Situation                  | Handling                                                                                                |
|--------------------|----------------------------|---------------------------------------------------------------------------------------------------------|
| <b>New learner</b> | No history                 | Tasters create signal in the first session; skill check supplies level; segment trending covers the gap |
| <b>New course</b>  | No enrolments              | 15% of slots reserved, chosen by content match. Graduates at 50 completions                             |
| <b>Permanent</b>   | Under-18, limited tracking | Never builds usable behavioural history. Served indefinitely by three methods                           |

The third is not a phase the learner grows out of. It is a permanent mode for a third of the main persona.

### Feedback loop

Clicks and enrolments feed the scoring model · completion and dropout adjust how highly a course ranks · fit survey answers score the recommender separately from course quality · new-course conversion tunes the 15% reservation.

## Step 5 · Trade-off analysis

### Tensions considered

| Tension                                                       | Verdict                                                                    |
|---------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Personalisation depth vs. protecting under-18 learners</b> | <b>Chosen</b>                                                              |
| New-course exposure vs. completion rate                       | Real, but it has a dial — the 15% reservation — making it a tuning problem |
| Engagement optimisation vs. genuine learning                  | Real, but already answered by the placement-rate guardrail                 |

The first has no clean answer, was imposed rather than chosen, and changed the architecture. The other two have dials.

### What's in conflict

**The learners who know themselves least are the ones we're least allowed to learn about.**

About a third of users are under 18, and Indian law limits tracking of minors. But a recommendation system is tracking. The product's premise — that a school leaver cannot declare a goal, so the system must infer one from behaviour — collides with the rule against inferring things from a minor's behaviour.

**Personalisation wants behaviour.** What people do is more truthful than what they say, and it is the only method that surfaces a match the learner would never have searched for. That is the track explorer's founding principle, and it is what becomes unavailable for minors.

**Protection wants restraint.** Legally: consent, limited retention, restricted use. Beyond law: a 17-year-old making a major career decision under the influence of a system they cannot inspect.

**Why not both.** The constraint is permission, not money or engineering. And it bites hardest where the value is highest — the adult career switcher who needs inference least is fully trackable; the 17-year-old who needs it most is not.

| Choice              | Cost falls on                                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Track everyone      | Legal exposure and parental trust, with parents being the payers; and minors nudged by inference they cannot examine           |
| Restrict all minors | The under-18 learner gets worse recommendations than the 19-year-old beside them. Same price, worse product, no way of knowing |

The second is a fairness problem as well as a quality one: a two-tier product where the disadvantaged tier cannot see it.

### How we'd prioritise

**One mistake is fixable. The other is not.**

Restricting too much costs recommendation quality — gradual, measurable, reversible. Restricting too little risks regulatory action, parental trust and harm to minors, which arrive suddenly and cannot be undone. The question is not which is bigger on average but which failure we can survive.

Restriction is also temporary per person. A 17-year-old turns 18. The group is permanent; nobody stays in it.

### The decision

**Restrict by default, offer a consented middle tier, and build the restricted tier to work without behavioural history.**

| Tier | Who                                   | Methods                                       | Retention         |
|------|---------------------------------------|-----------------------------------------------|-------------------|
| 1    | Under 18, processing consent declined | Three                                         | Minimum necessary |
| 2    | Under 18, processing consent given    | Four, recommendations only, never advertising | Shortened         |
| 3    | 18 and over                           | Four                                          | Standard          |

The tiering follows from the law. What does not follow from the law is the quality of tier 1, and a tier 1 account pays full price. So the taster explorer and the direct skill check are built to produce usable signal within one session, without accumulated history. That is why the architecture has an onboarding layer at all, and why signal freshness is 15 minutes.

### What we're giving up

1. **Fewer surprising discoveries for tier 1.** The remaining methods mostly confirm what the learner already indicated.
2. **Longer cold start** for the group that can least afford it.
3. **A harder business case.** If tier 1 underperforms, blended figures drag the targets down.

### Reducing the damage

1. **Measure the gap.** Report fit and completion separately for tier 1 and tier 3. If the gap crosses a threshold, that is a product failure to fix, not a compliance cost to absorb.
2. **Treat the track explorer as a primary signal source,** not a stopgap. Improvements there benefit tier 1 most.
3. **Design the consent flow as value to the parent** — progress visibility, outcome reporting — rather than friction to be minimised.
4. **Tell the learner.** An acknowledged two-tier product is better than a hidden one.

**What would change our mind**

- The tier 1 versus tier 3 gap turns out small
- Regulation clarifies that educational recommendations are exempt
- Consent uptake exceeds 70%, shrinking the problem to an edge case
- Tier 1 completion falls far enough that the fairness harm outweighs the tracking harm

## Step 6 · Assumptions and future planning

### Assumptions

| #  | Assumption                                                                                         | If wrong                                                                                         |
|----|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 1  | Paid, AI-first job-readiness platform in India                                                     | A different country changes all of Step 5                                                        |
| 2  | No personalisation exists today                                                                    | Baselines are wrong, and we would need to explain why the existing system failed                 |
| 3  | Two formats: self-paced and 8-16 week cohort programs                                              | A single format collapses the engagement weighting                                               |
| 4  | 12th-pass primary, career switchers secondary, hobbyists excluded                                  | If the main user has a goal already, the track explorer is unnecessary                           |
| 5  | About 35% of learners are under 18                                                                 | Sizes the Step 5 trade-off. At 5% it is a footnote; at 60% it should have shaped ranking further |
| 6  | 500,000 registered · 120,000 active · 800 courses · 250 instructors                                | An order-of-magnitude error invalidates most of Step 3                                           |
| 7  | 15,000 peak concurrent, 6,000 in live classes                                                      | A higher peak ratio forces earlier re-architecture                                               |
| 8  | Baselines of 42% engagement, 22% completion, 68% satisfaction                                      | All three targets are relative to invented numbers                                               |
| 9  | Satisfaction measured as share rating 4+, not a raw average                                        | If a raw average is meant, the target must be restated in points                                 |
| 10 | Completion defined by capstone, separate from engagement                                           | If it must cover non-capstone courses, a second tier is needed                                   |
| 11 | Placement rate obtainable, self-reported, six-month lag                                            | Without it, no defence against optimising proxies                                                |
| 12 | <b>Indian data protection law imposes consent and tracking duties for under-18s. Not verified.</b> | Step 5 depends entirely on this                                                                  |
| 13 | Consent permits limited processing for education but not advertising                               | Without a middle tier the decision becomes all-or-nothing                                        |
| 14 | Career switchers get no dedicated stories in version 1                                             | Acknowledged gap, in the roadmap                                                                 |

**Design parameters, all invented and adjustable:** 12 career tracks · 10-minute tasters, minimum three sampled · 8-12 test questions, 90-day retake · time bands of <5, 5-10, 10-15, 15+ hours · 50 completions before observed data replaces estimates · 8-10 paths of 6-10 courses · 15% new-course reservation · 14-day inactivity trigger · shortlist capped at ~500.

**Threshold sources:** 200ms follows convention for above-the-fold content and the reasoning holds at 300ms. 15-minute freshness is derived from the explorer needing to work in one session. Three times concurrency headroom covers admission-season spikes. 10,000 courses assumes 250 instructors publish faster than learners join. 99.5% sits below payments because a stale recommendation is recoverable.

### Left out of version 1

| Left out                                      | Why                                                 |
|-----------------------------------------------|-----------------------------------------------------|
| Ranking on job outcomes                       | Needs six-month-lagged data that does not exist yet |
| Class group matching by pace                  | Named as a dropout cause, served by no feature      |
| Career-switcher onboarding                    | Secondary persona, scoped out                       |
| Login, payments, video, moderation, authoring | Platform services                                   |
| Real-time model updates                       | Scheduled retraining is enough at this scale        |

### Roadmap

#### Next 6-12 months

1. **Rank on job outcomes.** Add a fifth shortlisting method ranking on gap-to-employment rather than gap-to-syllabus, turning placement rate from something we audit into something we steer by.
2. **Fix the tier 1 gap,** driven by the measured difference. If material, this takes priority over the rest of the roadmap.
3. **Class group matching** by compatible pace and availability rather than start date alone.
4. **Career-switcher route.** Skip the explorer for learners arriving with a goal.

#### 12-24 months

5. **Employer demand signal.** Job-posting data showing which skills are actually being hired for, feeding ranking and catalog approval. Makes job-readiness measurable against the market rather than asserted from our own skills list.
6. **Learn the new-course reservation rate** per track from observed conversion, replacing the fixed 15%.

### Known limitations

1. **Learning paths and next-course recommendations overlap** on "what should I do next." Resolved with a precedence rule rather than structural separation.
2. **The legal assumption is unverified,** and Step 5 rests on it.

3. **Placement rate cannot steer within the measurement year.** Six-month lag makes it an after-the-fact check, not an operational control.